MAT 504

Week 4 Hands-on – When Can I Trust an MD Trajectory?

Computer Modelling in Materials Science

Objective. Week 3 focused on how molecular dynamics generates a trajectory. Week 4 asks the next question: **when can an observable calculated from that trajectory be trusted?** In this exercise, you will use a small Lennard–Jones system to compare NVE and NVT dynamics, separate equilibration from production, calculate an equilibrium average, examine the autocorrelation of a specific observable, and use block averaging to estimate uncertainty from correlated data.
The conceptual sequence is: the **ensemble** defines the thermodynamic conditions, **equilibration** removes strong memory of the arbitrary starting state, **production** supplies data for analysis, and **correlation-aware statistics** determine how much independent information the trajectory contains.

Computational setup

A starter Jupyter notebook is provided. Complete the indicated `TODO` cells and answer the short questions directly in the notebook. The low-level molecular-dynamics machinery is supplied; the purpose of this session is to understand and analyze the workflow rather than to implement a thermostat or MD engine from scratch.

We use the same small two-dimensional Lennard–Jones fluid as in Week 3, in reduced units,

$$\epsilon = \sigma = m = k_B = 1, \qquad U(r) = 4\left(r^{-12} - r^{-6}\right).$$

Periodic boundaries are handled with the minimum-image convention. Because this is a small teaching system, all particle pairs are evaluated directly.

For NVT, the supplied helper uses a simple **Andersen thermostat**. At each integration step, each particle has a small probability of receiving a new velocity drawn from the target thermal distribution. You are not expected to derive or modify this thermostat.

Suggested pacing for a 60-minute session: NVE/NVT comparison (15 min), equilibration and production (15 min), production average (8 min), autocorrelation (10 min), block averaging and synthesis (12 min).

Part 1 – NVE versus NVT

Run two simulations from the same initial coordinates and velocities:

- **NVE:** no thermostat; total energy should remain approximately constant.
- **NVT:** target temperature $T = 1.0$; the thermostat exchanges energy with the system.

Both simulations start deliberately at $T = 0.5$, so the NVT relaxation toward its target is visible.

1.1 Predict before running

Answer briefly:

a. In which ensemble should total energy remain approximately constant?

b. In which ensemble should the temperature relax toward a chosen target?

c. In NVT, should the instantaneous temperature equal the target value exactly at every stored frame?

1.2 Compare the two runs

Plot, for both NVE and NVT:

- temperature versus time;
- total energy per particle versus time.

Then answer:

a. What indicates that the NVE trajectory behaves as expected for conservative dynamics?

b. Why is total energy not expected to remain constant in NVT?

c. Does the NVT temperature fluctuate around the target, or is it fixed exactly at the target?

Part 2 – Equilibration versus production

For the NVT trajectory, use

$$t_{\mathrm{eq}} = 10$$

as the proposed end of equilibration.

Plot:

- temperature versus time;
- potential energy per particle U/N versus time;

and mark t_{eq} on both plots.

> A single flat-looking curve is not proof of equilibration. For this short exercise, the question is only whether $t_{\mathrm{eq}} = 10$ is a reasonable working boundary based on the two observables shown.

Answer: what changes before and after $t = 10$, and why is it useful to exclude the early relaxation period from production statistics?

Part 3 – Calculate a production average

Define the scalar observable

$$A(t) = \frac{U(t)}{N}.$$

Calculate:

1. the mean U/N using the entire NVT trajectory;
2. the mean U/N using only the production region $t \geq 10$.

Explain why the production-only estimate is scientifically preferable even if the two numerical values are fairly close in this particular run.

Part 4 – Autocorrelation of a specific observable

Trajectory frames generated close together in time are not statistically independent. Here, the autocorrelation is explicitly the autocorrelation of

$$A(t) = U(t)/N.$$

Use the supplied `normalized_autocorrelation` function and plot $C_A(k)$ against lag time for the production data.

Answer:

a. What specific quantity is being autocorrelated?

b. What does decay of $C_A(k)$ tell you about the independence of neighboring stored values?

c. Why does saving frames more frequently not necessarily provide proportionally more independent information?

Part 5 – Block averaging and uncertainty

Divide the production U/N time series into **10 equal contiguous blocks**. Let the mean in block b be $\bar{A}_b$. The overall production mean can be estimated from the block means,

$$\bar{A} = \frac{1}{B}\sum_{b=1}^{B} \bar{A}_b.$$

Compare two standard-error estimates:

$$\mathrm{SE}_{\mathrm{naive}} = \frac{s_{\mathrm{frames}}}{\sqrt{N_{\mathrm{frames}}}},$$

$$\mathrm{SE}_{\mathrm{block}} = \frac{s_{\mathrm{blocks}}}{\sqrt{B}}.$$

The naive estimate treats every stored production frame as independent. The block estimate instead uses the spread among block means.

Plot the 10 block means and report both uncertainty estimates.

Answer:

a. Which uncertainty estimate is larger?

b. Why does ignoring time correlation tend to underestimate uncertainty?

c. Why should the block duration be long compared with the correlation time?

Final synthesis

In 3–4 sentences, explain why the following statement is incomplete:

> "My MD simulation ran without crashing and I saved many frames, so the reported average is trustworthy."

Your answer should mention **ensemble**, **equilibration/production**, **correlation**, and **uncertainty or convergence**.

Before you leave

Make sure your notebook contains:

1. your NVE/NVT prediction;
2. the NVE/NVT temperature and total-energy comparison;
3. the NVT equilibration plots;

4. the full-trajectory and production-only averages of U/N;
5. the autocorrelation of U/N;
6. the block means and two uncertainty estimates;
7. all written answers and the final synthesis.